25901232 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEECT203 - SINGLE BOARD COMPUTERS FOR IOT APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

MCQ

Answer all questions

Each question carries **1** marks

1. Drivers are required to _____. [K] / [CO1]

   a). Power Arduino b). Connect Arduino to computer c). Store programs d). Run sensors

2. Pull-up resistors are used to avoid _____. [K] / [CO1]

   a). Short circuit b). Floating input c). High current d). Overheating

3. IR sensor consists of _____. [U] / [CO1]

   a). LED only b). Photodiode only c). Transmitter and receiver d). Battery

4. Which language is used to program Arduino? [U] / [CO1]

   a). Java b). Python c). Embedded C d). Assembly

5. Heat conversion is an example of [K] / [CO2]

a). Conditional statement | b). Mathematical calculation | c). Looping | d). File handling

6. Ubuntu on Raspberry Pi supports [U] / [CO2]

a). ARM processor | b). Intel only | c). AMD only | d). 8051

7. What does an IoT cloud template define? [K] / [CO2]

a). Hardware wiring | b). Device properties and data model | c). Power supply | d). Sensor type

8. The area of a rectangle is calculated by [K] / [CO2]

a). Adding length and breadth | b). Multiplying length and breadth | c). Dividing length by breadth | d). Subtracting breadth from length

9. Arduino can interface directly with _____. [K] / [CO2]

a). Sensors and actuators | b). Printers only | c). Monitors only | d). Routers

10. ESP8266 can be programmed using _____ IDE. [K] / [CO2]

a). MATLAB | b). Arduino IDE | c). Photoshop | d). Tally

11. A variable in Python is created when [K] / [CO2]

a). Declared with type | b). Assigned a value | c). Compiled | d). Installed

12. An LED connected to GPIO acts as [K] / [CO2]

a). Input device | b). Output device | c). Storage device | d). Network device

13. BeagleBone supports which operating system? [U] / [CO3]

a). Windows XP | b). Linux | c). DOS | d). macOS only

14. Real-time monitoring means [K] / [CO3]

a). Delayed data processing | b). Instant data processing | c). Manual recording | d). Offline storage only

15. OpenCV is commonly used in [K] / [CO3]

a). Computer vision applications | b). Word processing | c). Email writing | d). File compression

16. Jetson Nano is powered by which GPU architecture? [U] / [CO3]

a). Pascal | b). Volta | c). Maxwell | d). Turing

[1x16 = 16]

Part B

Short Answer Questions

Answer any **6** questions

Each question carries **5** marks

17. What is digital output in Arduino? [K] / [CO1]

18. Mention any two features of Blynk. [U] / [CO1]

19. Define Internet of Things (IoT). [K] / [CO1]

20. What is Raspbian OS? [U] / [CO2]

21. Briefly explain the evolution of Raspberry Pi models [K] / [CO2]

22. What is the function of the SD card slot in Raspberry Pi? [U] / [CO2]

23. Mention any two applications of camera interfacing using OpenCV. [K] / [CO3]

24. State any two applications of Banana Pi. [U] / [CO3]

[5x6 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **12** marks

25. Compare Arduino Uno and Arduino Mega with respect to memory, I/O pins, and applications. [U] / [CO1]

26. Explain different Raspberry Pi models and their specifications. [K] / [CO2]

27. Discuss the importance of Linux commands in Raspberry Pi operations. [K] / [CO2]

28. Discuss the applications of Tinkerboard in embedded and IoT systems. [U] / [CO3]

[12x2 = 24]